

Enthusiast, Leader & Achiever Course

PHASE : (All Phase)

TARGET : PRE-MEDICAL 2020

Test Type : MAJOR

Test Pattern : NEET (UG)

TEST DATE : 07-05-2020

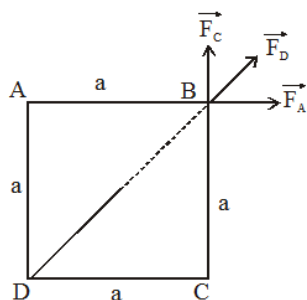
Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	3	2	3	3	1	2	2	1	2	1	1	3	2	3	3	4	1	2	4	1	1	4	4	4	2	1	2	3	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	3	1	1	1	2	3	1	1	2	3	2	3	3	2	4	3	4	4	4	3	3	3	2	4	1	2	1	4	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	2	1	2	2	3	2	2	4	4	2	2	3	3	1	4	3	1	4	1	2	1	2	3	2	3	4	4	1	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	3	3	1	4	1	1	4	2	1	1	4	1	4	3	3	4	4	4	2	2	2	2	1	2	3	3	1	2	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	4	3	2	1	4	2	2	2	2	3	3	2	2	4	1	4	1	3	1	1	2	4	4	1	3	1	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	1	2	1	1	2	2	2	2	2	3	1	1	3	4	3	4	1	1	4	4	2	1	3	2	3	4	4	2	1

HINT - SHEET

1. Ans (1)

All statements are correct.

2. Ans (3)



Total force on charge at B is

$$\vec{F}_B = \vec{F}_A + \vec{F}_C + \vec{F}_D$$

$$|\vec{F}_A| = |\vec{F}_C| = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a^2}$$

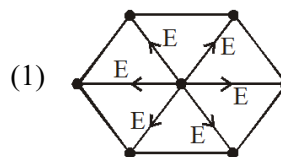
$$|\vec{F}_D| = \frac{1}{4\pi\epsilon_0} \frac{q^2}{(a\sqrt{2})^2}$$

$$|\vec{F}_A + \vec{F}_C| = \frac{\sqrt{2}q^2}{4\pi\epsilon_0 a^2}$$

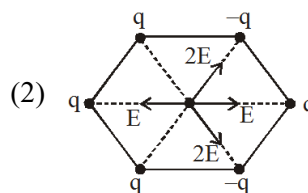
$\vec{F}_A + \vec{F}_C$ is along $\vec{F}_D$

$$\text{so } F_B = \frac{\sqrt{2}q^2}{4\pi\epsilon_0 a^2} + \frac{q^2}{4\pi\epsilon_0 (2a^2)}$$

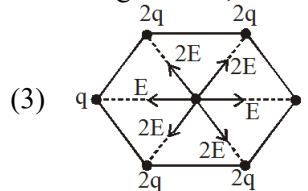
3. Ans (2)



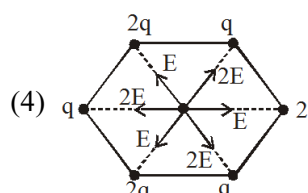
Sum of all electric fields is zero.



Sum of all electric fields is rightwards, non-zero

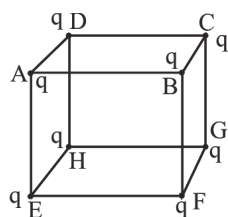


Sum of all electric fields is zero



Sum of all the electric fields is zero

4. **Ans (3)**



By symmetry ϕ due to charge at E, F, G & H from face ABCD are equal

$$\phi \text{ due to charge at E} = \left(\frac{q}{8}\right) \times \frac{1}{\epsilon_0} \times \frac{1}{3}$$

$\swarrow$ q_{in} $\searrow$ 3 face

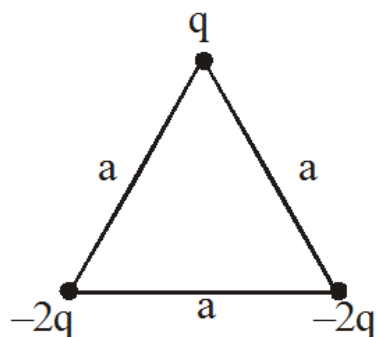
$$\phi \text{ due to charge at E, F, G, H} = 4 \times \frac{q}{24\epsilon_0} = \frac{q}{6\epsilon_0}$$

ϕ due to charge at A, B, C & D = 0 as field lines are perpendicular to area vector

5. **Ans (3)**

$V_+ = -(V_-)$ at the axial position. PE of system is zero

6. **Ans (1)**



Initial potential energy

$$U_i = \frac{kq(-2q)}{a} + \frac{kq(-2q)}{a} + \frac{k(-2q)(-2q)}{a}$$

$$U_i = 0$$

If side of triangle become 2a

$$U_f = 0 \text{ also so } W = U_f - U_i = 0$$

7. **Ans (2)**

$$v = 0 + \left(\frac{qE}{m}\right)t$$

$$\text{and KE} = \frac{1}{2}mv^2 = \frac{q^2E^2t^2}{2m}$$

8. **Ans (2)**

$$Q = q_1 + q_2 = 1 \text{ mC}, \quad q_1 : q_2 :: 2 : 3$$

$$\text{let } q_1 = 2q, \quad q_2 = 3q, \quad Q = 2q + 3q = 5q$$

$$q = \frac{1}{5}\mu\text{C} = 0.2\mu\text{C}$$

$$F = \frac{kq_1q_2}{r^2} = \frac{9 \times 10^9 \times (2 \times 0.2 \times 10^{-6})(3 \times 0.2 \times 10^{-6})}{1^2}$$

$$F = 0.00216 \text{ N}$$

9. **Ans (1)**

$$V_{in} =$$

$$\frac{kq_{in}}{r_{in}} + \frac{kq_{out}}{r_{out}}, \quad V_{out} = \frac{kq_{in}}{r_{out}} + \frac{kq_{out}}{r_{out}}$$

$$\text{potential difference } V_{in} - V_{out} = \frac{kq_{in}}{r_{in}} - \frac{kq_{in}}{r_{out}}$$

$V_{in} - V_{out}$ does not depend on outer charge

so $V_{in} - V_{out}$ remain unchanged = V

10. **Ans (2)**

$$\text{Between plate E} = \frac{\sigma}{2\epsilon_0} - \left(\frac{-\sigma}{2\epsilon_0}\right) = \frac{\sigma}{\epsilon_0},$$

$$\text{outside plate } \sigma = 0 \quad E' = 0.$$

11. **Ans (1)**

$$q = Ce(1 - e^{-t/RC})$$

$$V = e(1 - e^{-t/RC})$$

$$= \frac{q}{C} \text{ so straight line}$$

12. **Ans (1)**

$$E_I = \frac{\sigma}{3\epsilon_0}, \quad E_{II} = \frac{\sigma}{6\epsilon_0}$$

$$U_I = \frac{1}{2}\epsilon_0(3)E_I^2, \quad U_{II} = \frac{1}{2}\epsilon_0(6)E_{II}^2$$

$$\frac{U_I}{U_{II}} = \frac{1}{2} \times 2^2 = 2$$

13. **Ans (3)**

Energy density = u

$$u = \frac{1}{2}E_0E^2, \quad \text{if } E = \frac{\sigma}{\epsilon_0}$$

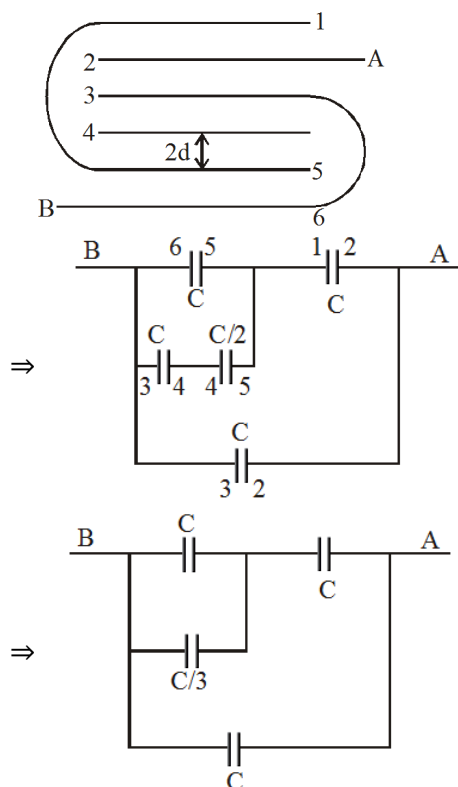
$$u = \frac{1}{2}\epsilon_0\left(\frac{\sigma}{\epsilon_0}\right)^2 = \frac{\sigma^2}{2\epsilon_0}$$

$$\text{If } E = \frac{v}{d}, \quad u = \frac{1}{2}\epsilon_0\left(\frac{v}{d}\right)^2$$

$$\sigma = \frac{Q}{A} \Rightarrow u = \left(\frac{Q}{A}\right) \frac{2}{2E} = \frac{Q^2}{2A^2\epsilon_0}$$

option (3) is not energy density

14. Ans (2)



$$C = \frac{\epsilon_0 A}{d} = 7 \mu\text{F}$$

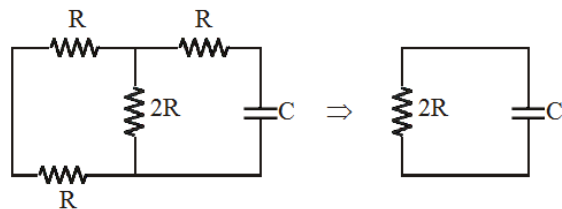
$$C_{AB} = C + \frac{\frac{4C}{3} \times C}{\frac{4C}{3} + C}$$

$$= C + \frac{4C}{7} = \frac{11C}{7}$$

$$C_{AB} = 11 \mu\text{F}$$

15. Ans (3)

On short circuit of battery



Time constant $\tau = 2RC$

16. Ans (3)

In series charge flow Q_1 through battery accumulate on C_2 and C_3

$$\text{so } Q_1 = Q_2 + Q_3$$

In parallel potential remain same

$$V_2 = V_3, V = V_1 + V_2$$

17. Ans (4)

$$Q = I t = 5 \times 60 = 300$$

18. Ans (1)

For Ist case

efficiency =

$$\frac{\text{TPD}}{\text{EMF}} \times 100 = \frac{ER/R + r}{E} = \frac{R}{R + r}$$

$$0.6 = \frac{R}{R + r}$$

$$\text{Hence } r = \frac{2}{3}R$$

In 2nd case

$$x = \frac{6R}{6R + r}$$

Put the value of r and get the answer.

19. Ans (2)

To obtain minimum resistance, all resistors must be connected in parallel.

Hence equivalent resistance of combination

$$\frac{r}{10}$$

20. Ans (4)

At steady state terminal potential difference between battery

$$V = IR = \frac{ER}{r + R} = \frac{2.5 \times 2}{0.5 + 2} = 2V$$

Charge on capacitor $Q = CV = 4 \times 2 \mu\text{C}$.

21. Ans (1)

By theory, internal resistance of ideal battery is zero, for constant current source resistance is high so no effect of circuit resistance.

22. **Ans (1)**

I case

$$\frac{10}{R} = \frac{\ell}{100 - \ell} \quad \dots (i)$$

$$\text{II case } \frac{10}{2R} = \frac{3/4 \ell}{100 - 3/4 \ell} \quad \dots (ii)$$

Form (i) and (ii)

$$2 = \frac{4}{3} \frac{100 - 3/4 \ell}{100 - \ell} \Rightarrow \ell = \frac{200}{3}$$

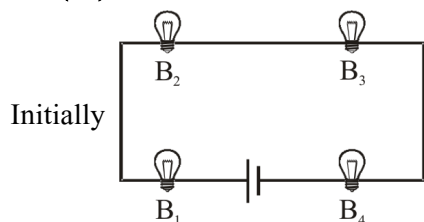
Put the value of ℓ in equation (i)

$$\frac{10}{R} = \frac{200/3}{100 - \frac{200}{3}} = 2 \Rightarrow R = 5\Omega$$

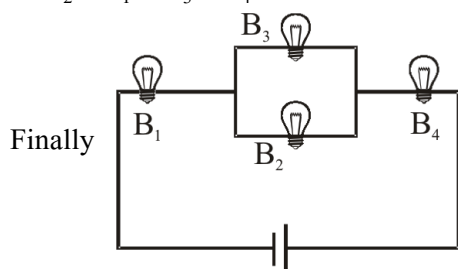
24. **Ans (4)**

Room heater require wire of high resistivity, convert electrical energy into heat, so Nichrome is preferred it convert electric energy into light energy.

25. **Ans (4)**



$$\Rightarrow R_2 > R_1 > R_3 = R_4$$



$$\Rightarrow \text{Order of brightness } B_1 > B_4 > B_3 > B_2$$

26. **Ans (2)**

$$F = \frac{1}{4\pi\epsilon_0} \frac{(2e)(2e)}{(8.2 \times 10^{-15})^2} = 14 \text{ N}$$

27. **Ans (1)**

Net electric field at O is zero.

$$\text{so } \vec{E}_{AKB} + \vec{E}_{\text{remaining}} = \vec{0}$$

$$\vec{E}_{\text{remaining}} = -\vec{E}_{AKB} = E \text{ along OK}$$

28. **Ans (2)**

By theory

29. **Ans (3)**

$$V_0 = \frac{k(10 + 5 - 3 + 8)}{r}$$

$$k = 9 \times 10^9, r = 1\text{m}$$

$$V_0 = 9 \times 10^9 \times 20 \times 10^{-16}$$

$$V_0 = 1.8 \times 10^5 \text{ V}$$

30. **Ans (2)**

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} = k, \text{ here } k = \frac{q}{4\pi\epsilon_0}$$

$$\vec{E} = \frac{q\vec{r}}{4\pi\epsilon_0 r^3} = \frac{(2\hat{i} + 3\hat{j} + 6\hat{k}) k}{(2^2 + 3^2 + 6^2)^{3/2}}$$

$$\vec{E} = \frac{(2\hat{i} + 3\hat{j} + 6\hat{k}) k}{343}$$

31. **Ans (1)**

q_1 has more deflected so more curved trajectory

32. **Ans (3)**

$$\vec{E}_{\text{equator}} = -\frac{K\vec{p}}{r^3}$$

Angle between $\vec{E}$ and $\vec{p}$ at equator is 180° .

33. **Ans (1)**

The distribution of charge on the outer surface, depends only on the charges outside, and it distributes itself such that the net, electric field inside the outer surface due to the charge on outer surface and all the outer charges is zero. Similarly the distribution of charge on the inner surface, depends only on the charges inside the inner surface, and it distributes itself such that the net, electric field outside the inner surface due to the charge on inner surface and all the inner charges is zero.

Also the force on charge inside the cavity is due to the charge on the inner surface. Hence answer is option (1).

34. **Ans (1)**

$$C = C_1 + C_2 = \frac{k_1 \epsilon_0 A/2}{d} + \frac{k_2 \epsilon_0 A/2}{d}$$

$$C = \frac{A \epsilon_0 (k_1 + k_2)}{2d}$$

35. **Ans (1)**

$$U = \frac{q^2}{2C}, \text{ dielectric is inserted then}$$

$$C \uparrow \quad U \downarrow$$

36. **Ans (2)**

As alternative plates are connected to the left and right side of the battery, four capacitors are connected in parallel combination and charge across each is

$$q = CV$$

Two surfaces of plate C marks two capacitors, one with B and other with D and C is connected to positive terminal of the battery so total charge on plate C is given as

$$q_C = 2CV = +40\mu C$$

37. **Ans (3)**

Charge flow in series is same on all three capacitors.

$$q = C_{eq} V = \frac{2}{3} \times 10^{-6} \times 100 = \frac{2}{3} \times 10^{-4} C$$

38. **Ans (1)**

$$\text{Current density } J = ar^2$$

$$\text{Current } I = \int J dA$$

$$I = \int_{R/3}^{R/2} ar^2 (2\pi r dr), \quad [\text{where } dA = 2\pi r dr]$$

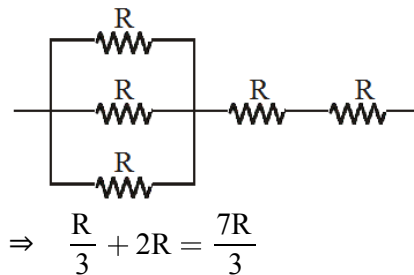
$$= 2\pi a \left[\frac{r^4}{4} \right]_{R/3}^{R/2} = \frac{65\pi a R^4}{2592}$$

39. **Ans (1)**

$$R \propto \frac{\ell}{A} \text{ on stretching } A\ell = \text{constant}$$

$$R \propto \ell^2 \Rightarrow R^1 = (1.1)^2 \times 10$$

40. **Ans (2)**



41. **Ans (3)**

$$\text{By KCl } 4 + I + 2 - 5 - 3 = 0$$

$$I = 2 \text{ amp.}$$

42. **Ans (2)**

$$I = \frac{E_1 + E_2 + E_3 + \dots + E_n}{r_1 + r_2 + r_3 + \dots + r_n} = \frac{k(r_1 + r_2 + r_3 + \dots + r_n)}{r_1 + r_2 + r_3 + \dots + r_n}$$

$$I = K$$

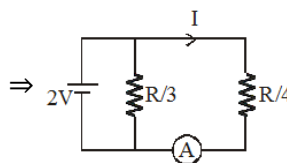
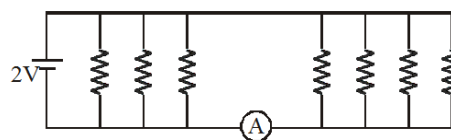
43. **Ans (3)**

Given circuit is balanced wheatstone bridge

$$R_{eq} = 6\Omega, I = \frac{2}{6} = 0.33A$$

44. **Ans (3)**

Let reading of ammeter is I



$$I = \frac{2V}{R/4} = \frac{2}{20/4} = \frac{4}{10} A$$

45. Ans (2)

$$30 = \frac{V^2}{R_1}, \quad 60 = \frac{V^2}{R_2}, \quad \frac{R_1}{R_2} = 2$$

In series $V = V_1 + V_2 = IR_1 + IR_2$

$$\frac{V_1}{V_2} = \frac{IR_1}{IR_2} = 2$$

47. Ans (3)

In ZnS (Zinc Blend)



$Zn^{+2} \rightarrow$ alternate (50%) THV so total 4 Zn^{+2} present per unit cell.

53. Ans (3)

In lyophilic sols, dispersed phase has affinity towards dispersion medium lyophilic sols are more stable than lyophobic sols because colloidal particles are solvated.

57. Ans (2)

If $E_a = 0$

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\log \frac{k_2}{k_1} = 0$$

then $k_2 = k_1$

63. Ans (1)

In Schottky defects and ions are missing from their position and in Frenkel defect ions are shifted in their interstitial site from their lattice point.

67. Ans (2)

$$r = K[B]$$

$$0.1 = K \times .5$$

$$K = 1/5 = .2$$

$$0.4 = K y$$

$$y = 2$$

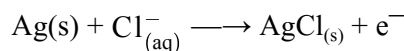
70. Ans (4)

$$(t_{1/2}) \propto \frac{1}{(A_0)}$$

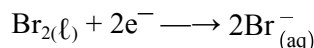
71. Ans (2)

For given cell

Anode reaction :



Cathode reaction



75. Ans (1)

$$\begin{aligned} \Lambda_{eq}^\infty Al_2(SO_4)_3 &= \Lambda_{eq}^\infty (Al^{+3}) + \lambda_{eq}^\infty (SO_4^{-2}) \\ &= 189 + 160 \\ &= 349 \text{ ohm}^{-1} \text{ cm}^2 \text{eq}^{-1} \end{aligned}$$

80. Ans (1)

$$\begin{aligned} E_{cell}^0 &= \frac{0.059}{2} \log 10^{12} \\ &= \frac{0.059}{2} \times 12 = 0.059 \times 6 \end{aligned}$$

82. Ans (1)

$$\begin{aligned} P_g &= K_H \cdot \frac{n_g}{n_{H_2O}} \\ 1 &= K_H \cdot \frac{0.02}{28} \times \frac{18}{1000} \\ K_H &= \frac{28 \times 1000}{0.02 \times 18} = 7.7 \times 10^4 \text{ atm} \end{aligned}$$

84. Ans (3)

$$\text{Mili equivalent of HCl} = 100 \times 0.3 = 30$$

$$\text{Mili equivalent of H}_2\text{SO}_4 = 200 \times 0.6 = 120$$

$$\therefore N_{\text{mixture}} = \frac{30 + 120}{300} = \frac{1}{2}$$

86. Ans (3)

$$n_{\text{glucose}} = \frac{10}{180}, \quad n_{\text{urea}} = \frac{5}{60}$$

$$n_{\text{KCl}} = 2 \times \frac{1}{74.5}, \quad (\text{electrolyte } i = 2)$$

$$\begin{aligned} \therefore \Delta T_f &= K_f \times m = \frac{1.86 \times \left(\frac{10}{180} + \frac{5}{10} + \frac{2}{74.5} \right)}{\frac{84}{1000}} \\ &= 3.7 \end{aligned}$$

87. **Ans (4)**

$$1 + 2\alpha = \frac{165}{65} = 2.53$$

$$2\alpha = 1.53$$

$$\alpha = \frac{1.53}{2} = 0.75$$

$$\% \alpha = 75\%$$

 88. **Ans (4)**

$$i = 1 + 3\alpha \Rightarrow 1 \times 3 \times 0.9 \Rightarrow 3.7 ;$$

$$\Delta T_b = iK_b \cdot m \Rightarrow 0.52 \times 0.1 \times 3.7 \Rightarrow 0.19$$

$$\therefore T_b = T_b^\circ + 0.19 \Rightarrow 373 + 0.19 \Rightarrow 373.19$$

 89. **Ans (1)**

$$(P_s)_{cal.} = P_A^\circ X_A + P_B^\circ X_B$$

$$= \frac{1}{3} \cdot (150) + \frac{2}{3} (300)$$

$$= 50 + 200$$

$$= 250$$

$$\therefore (P_s)_{obs} < (P_s)_{cal.}$$

$\therefore$ Negative deviation

 90. **Ans (1)**

Liquid A has lower boiling point hence will have higher vapour pressure.

 92. **Ans (3)**

NCERT Pg# 127

 97. **Ans (1)**

Module-3 Pg. # 115

 101. **Ans (1)**

NCERT XII (E) Pg # 129, (H) Pg # 139

 103. **Ans (1)**

NCERT (XII) Pg # 140/149(H) Para 7.8

 113. **Ans (2)**

NCERT-XII, Pg. # 132

 116. **Ans (3)**

NCERT (XIIth) Pg. # 140,141

 125. **Ans (2)**

NCERT (XIIth) Pg. # 47

 128. **Ans (2)**

NCERT XII Pg. # 49

 132. **Ans (3)**

NCERT Pg.# 54, Para. 3.6

 137. **Ans (1)**

NCERT (XII) Pg. # 54, para 3

 142. **Ans (1)**

NCERT (XIIth) Pg. # 61 (E) 68 (H)

 143. **Ans (2)**

NCERT (XIIth) Pg. # 64

 151. **Ans (4)**

NCERT-XII, Page#21

 153. **Ans (2)**

NCERT XII Page # 26

 157. **Ans (2)**

1 M.M.C = 4 Pollen Grain

100 M.M.C = $100 \times 4 = 400$ Pollen Grain

 159. **Ans (2)**

NCERT (XII) Page # 26,27

 162. **Ans (1)**

NCERT (XII) Pg. # 27, 2.2.3

 164. **Ans (3)**

NCERT XII Pg 34(E), 35-37(H)

A and C are wrong

 169. **Ans (1)**

NCERT XII Pg # 35

 174. **Ans (3)**

NCERT XII Eng Pg # 38 para 6 line 8,9

 177. **Ans (4)**

Middle layer is 2N

 178. **Ans (4)**

Module-1, Pg # 115-116(E), 116(H)

 180. **Ans (1)**

NCERT (XIIth) Pg. # 7